VISITOR MANAGEMENT MODULE (CAMPUS ACCESS & SECURITY ENGINE)

SmartCampus School Management System
Technical Specification & Implementation Guide — v1.0 • March 2026
Confidential • Internal Documentation

Audience: School Administrators • Security Personnel • IT Team • Parents • Visitors
Related Modules: Student Management • Transport • Hostel • Notifications • Security • Parent Portal

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1. MODULE OVERVIEW

1.1 Purpose

The Visitor Management Module controls and monitors all external access to the school premises. It ensures security, accountability, and compliance by enabling:

- Visitor registration and check-in/check-out tracking
- Pickup authorization and logging
- Gate and access point monitoring
- Real-time visibility of all visitors on campus
- Integration with security systems and notifications

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2. CORE OBJECTIVES

1. Track all visitors entering and leaving the school in real time
2. Ensure secure authentication and verification of visitors
3. Manage student pickups with proper authorization
4. Provide visibility to security personnel and administrators
5. Enable emergency tracking of all persons on campus
6. Integrate with CCTV, student records, and notification systems
7. Support mobile-first and fast check-in/check-out processes
8. Ensure compliance with data protection and security policies

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3. REAL-WORLD SYSTEM FLOW

3.1 Lifecycle Flow

Visitor Arrives → Identity Verified → Check-In Logged → Access Granted → Visit Monitored → Check-Out Logged → Reports Generated

3.2 Daily Operational Flow

Visitor arrives at gate
↓
Security verifies identity (ID/QR/badge)
↓
Visitor checked into system
↓
Access granted to specific location/person
↓
Visitor monitored during stay
↓
Visitor checks out
↓
Logs updated and analytics generated

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4. MODULE ARCHITECTURE

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Visitor Management Module
│
├── Visitor Registration & Check-In
├── Identity Verification & Badge Issuance
├── Pickup Authorization System
├── Gate & Access Point Monitoring
├── Real-Time Tracking Dashboard
├── Emergency & Evacuation Tracking
├── Notifications & Alerts
├── Analytics & Reporting
├── CCTV & Security Integration
├── Mobile & QR Code Access
├── Compliance & Data Protection
├── Role-Based Access Control
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5. VISITOR REGISTRATION & CHECK-IN

5.1 Features

- Log visitor details (name, role, purpose)
- Record check-in and check-out timestamps
- Assign visitor to host (student/staff)
- Maintain complete visitor logs

5.2 Data Model

Visitor

- visitor_id
- full_name
- role (parent/vendor/contractor/etc.)
- purpose
- host_id
- check_in_time
- check_out_time
- status

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6. IDENTITY VERIFICATION & BADGE ISSUANCE

6.1 Features

- ID scanning (national ID/passport)
- Temporary visitor badge generation
- QR code-based verification
- Optional integration with national ID systems

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7. PICKUP AUTHORIZATION SYSTEM

7.1 Features

- Verify authorized guardians for student pickup
- Log pickup events with timestamps
- Alert unauthorized pickup attempts

7.2 Data Model

PickupLog

- pickup_id
- student_id
- visitor_id
- authorized (true/false)
- time
- status

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8. GATE & ACCESS POINT MONITORING

8.1 Features

- Track multiple entry/exit points
- Monitor activity per gate
- Identify unmanned or low-security access points

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9. REAL-TIME TRACKING DASHBOARD

9.1 Features

- View all visitors currently on campus
- Monitor duration of visits
- Highlight overdue or suspicious presence

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10. EMERGENCY & EVACUATION TRACKING

10.1 Features

- Identify all visitors on campus during emergencies
- Support evacuation tracking
- Trigger alerts for unaccounted individuals

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11. NOTIFICATIONS & ALERTS

11.1 Features

- Alerts for unauthorized access attempts
- Notifications for visitor arrivals
- Pickup alerts for parents and staff
- Emergency alerts

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12. ANALYTICS & REPORTING

12.1 Features

- Visitor trends (daily, weekly, monthly)
- Peak visiting hours
- Frequent visitors and roles
- Security incident reports

12.2 Dashboards

- Visitor activity overview
- Gate usage analytics
- Pickup activity summaries

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13. CCTV & SECURITY INTEGRATION

13.1 Features

- Link visitor logs with CCTV feeds
- Real-time monitoring of entry points
- Event-based video playback

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14. MOBILE & QR CODE ACCESS

14.1 Features

- QR-based check-in/check-out
- Mobile app for security staff
- Fast scanning for high-traffic periods

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15. COMPLIANCE & DATA PROTECTION

15.1 Features

- Visitor consent forms
- Data retention policies
- Compliance with child protection and privacy laws

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16. SECURITY & ROLE-BASED ACCESS

Role	Access Scope
Admin	Full access, reports, configuration
Security Staff	Manage check-ins, monitor visitors
Teacher	View assigned visitors
Parent	Pickup authorization only

16.2 Security Features

- Secure authentication and session management
- Audit logs for all visitor activities
- Encrypted data storage and transmission

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17. SCALABILITY

- Multi-campus support with multiple access points
- High-volume visitor handling
- Cloud-based storage and backups

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18. IMPLEMENTATION GAPS (CRITICAL FIXES)

18.1 Current Issues Identified

- No ID verification or badge system
- Missing emergency tracking features
- Limited analytics and reporting
- Weak integration with other modules (SIS, CCTV)
- No mobile or QR-based fast check-in

18.2 Required Fixes

- Implement ID scanning and badge issuance
- Add emergency evacuation tracking
- Build analytics dashboards
- Integrate with student and security systems
- Enable mobile and QR-based check-in

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19. FUTURE ENHANCEMENTS

- Facial recognition for visitor identification
- AI-based anomaly detection (suspicious behavior)
- Smart gate automation (biometric access)
- Integration with national security systems
- Predictive visitor flow analytics